

MATLAB/Simulink Development Plan

Coupled Electro-Thermal-Pressure (ETP) Model for 21700 Cell Thermal Runaway

This is the engineering plan you will execute. The paper draft is the destination; this is the route. Work through phases in order — do not jump ahead, because each phase produces a result that the next phase depends on.

Phase 0 — Foundations (Week 1)

Goal: confirm you can do the modelling work, and obtain the validation data.

0.1 Get the dataset

Download the Gulsoy et al. 2025 dataset from the Data in Brief article:

- DOI: [10.1016/j.dib.2025.112190](https://doi.org/10.1016/j.dib.2025.112190)
- Published 15 October 2025
- Three thermal runaway tests, 21700 cells, with: cell voltage, internal gas pressure, surface temperature near both terminals, surface + internal temperature at axial midpoint, vent temperatures at 5 mm and 10 mm.

The supplementary data is hosted on the journal's site. If you cannot get it, the Battery Failure Databank (NASA/NREL — Iontech open-source list) has comparable thermal runaway data you can use as a fallback validation source. **Do not start any modelling before you have the validation data in hand.** A model without validation data is a creative-writing exercise.

0.2 Read these three papers properly, not the abstracts

1. **Hatchard et al. 2001** — the original reaction kinetics. You need to understand every equation in here. *J. Electrochem. Soc.* 148(7), A755.
2. **Kim et al. 2007** — three-dimensional thermal abuse extension. *J. Power Sources* 170(2), 476.
3. **Gulsoy et al. 2024** — the experimental methodology paper that produced your validation dataset. *J. Power Sources* 617, 235147.

Take handwritten notes on the reaction equations, activation energies, and pre-exponentials. You will need these as starting values in Phase 2.

0.3 Set up the workspace

- MATLAB R2023b or later (Simulink + Simscape Battery if available; Simscape Battery is *not* required for this work but helps with later visualisation).
- Create a Git repository on GitHub from day one. Every commit is dated evidence of independent work. This matters for the conversation with Prof. Marco.
- Recommended folder structure:

```
/etp_model
  /data          (Gulsoy dataset, parsed)
  /params        (parameter .mat files)
  /simulink      (.slx files)
  /scripts       (parameter ID, plotting)
  /results       (figures, tables)
  README.md
  LICENSE
```

Phase 1 — Reproduce a Known Model (Weeks 2-3)

Goal: before you build anything novel, reproduce a published TR model. This calibrates your understanding and gives you a baseline to extend.

1.1 Implement the Hatchard-Kim 0D model in Simulink

Build four parallel integrator chains, one per reaction. Reaction extents (state variables):

- x_{sei} — SEI fraction remaining, initial value 0.15 (Kim et al.)
- c_{ne} — anode reactive lithium fraction, initial 0.75
- a — cathode conversion, initial 0.04
- c_e — electrolyte fraction, initial 1.0

Each rate equation is:

$$dx/dt = A * f(x) * \exp(-E_a / (R * T))$$

Reaction-specific forms (refer to Kim Table 2):

Reaction	A (1/s)	Ea (J/mol)	f(x)	ΔH (J/kg)
SEI decomp	1.667e15	1.3508e5	x_{sei}	2.57e5
Anode-elec	2.5e13	1.3508e5	$c_{\text{ne}} \cdot \exp(-x_{\text{sei}}/0.033)$	1.714e6
Cathode decomp	6.667e13	1.396e5	$\alpha(1-\alpha)$	3.14e5
Electrolyte	5.14e25	2.74e5	c_e	1.55e5

⚠ These values are starting points only. They were fitted by Kim et al. to an LCO chemistry, so for NMC you will need to re-fit (Phase 2.2).

1.2 Add the thermal balance

Lumped energy balance, single Simulink integrator for T :

$$m * c_p * dT/dt = Q_{\text{react}} + Q_{\text{ext}} - h * A_s * (T - T_{\text{amb}}) - \epsilon * \sigma * A_s * (T^4 - T_{\text{amb}}^4)$$

For a 21700 NMC:

- $m \approx 0.069$ kg (depends on exact cell)
- $c_p \approx 1100$ J/(kg·K)
- $A_s \approx 0.0042$ m² (surface area)
- $h \approx 10\text{--}25$ W/(m²·K) (natural convection, identify from data)
- $\epsilon \approx 0.8$ (typical for casing)

Q_{react} sums the four reaction contributions: $Q_i = \Delta H_i \cdot m_{\text{active}_i} \cdot (dx_i/dt)$.

Q_{ext} is your time-series block matching the experimental heating pad profile (read this from the dataset's metadata).

1.3 Solver settings

- Solver: `ode15s` (stiff)
- Relative tolerance: `1e-6`

- Absolute tolerance: $1e-9$
- Max step size: 0.5 (s)

Anything looser misses the onset transient; anything tighter wastes time.

1.4 Sanity check

Run the model with $T(0) = 25\text{ }^{\circ}\text{C}$, $Q_{\text{ext}} = 30\text{ W}$ step input, ambient $25\text{ }^{\circ}\text{C}$. You should see:

- A slow temperature rise for ~10 minutes
- Sharp acceleration when T crosses $\sim 140\text{ }^{\circ}\text{C}$
- Peak temperature in the $600\text{--}800\text{ }^{\circ}\text{C}$ range
- Total simulated time under 30 minutes

If you do not see this shape, your reaction parameters are wrong or your Q_{react} sign is flipped. Do not proceed until the shape is right.

Phase 2 — Parameter Identification (Weeks 4–5)

Goal: fit the model to ONE cell from the Gulsoy dataset (the "training" cell), reserving the other two for validation.

2.1 Parse the dataset

Write `scripts/load_gulsoy_data.m` that returns a struct array with fields:

```
data(i).t           % time vector, s
data(i).T_internal  % internal axial-midpoint temperature, °C
data(i).T_surface_mid % surface midpoint temperature, °C
data(i).P_internal  % internal gas pressure, bar absolute
data(i).V_cell       % cell voltage, V
data(i).heater_power  % applied heater power, W (from metadata)
```

2.2 Re-fit reaction kinetics to NMC

The Kim parameters are LCO-fitted. For NMC, the cathode decomposition parameters need adjustment. Use MATLAB's `fmincon`:

matlab

% Pseudocode

```
params0 = [A_cath, Ea_cath, H_cath]; % cathode decomposition parameters
lb = params0 .* [0.3, 0.85, 0.7]; % stay within ±15% Ea, wider on A and H
ub = params0 .* [3.0, 1.15, 1.4];
opts = optimoptions('fmincon', 'Display', 'iter', 'MaxFunctionEvaluations', 1000);
params_fit = fmincon(@(p) objective(p, data(1)), params0, [], [], [], [], [], lb, ub, opts);
```

The `objective` function runs the Simulink model with the candidate parameters and returns the L2 norm of the temperature error against `data(1).T_internal`.

⚠ Do not let the optimiser overfit. Hard bounds on activation energies ($\pm 15\%$ of literature) are essential. A loosely-bounded optimiser will produce values that fit training data perfectly and have no physical meaning. Reviewers will catch this immediately.

2.3 Identify thermal loss coefficient

Before TR onset ($T < 100^\circ\text{C}$), the dynamics are dominated by heat input and loss, not reaction. Fit h and ε from the slow-rise pre-onset portion of the data, then freeze them.

Phase 3 — Add the Pressure Sub-Model (Weeks 6–7)

Goal: this is your novel contribution. Everything before this is reproduction.

3.1 Implement the gas generation integrator

Add a fifth integrator for n_g (moles of gas in the cell free volume V_{int}).

$$\frac{dn_g}{dt} = v_{vap} * k_{vap}(T) * m_e(t) + v_e * |dc_e/dt| * n_{e0} + v_{O2} * |da/dt| * n_{c0}$$

For a 21700 cell, $V_{int} \approx 1.5\text{--}2.5\text{ mL}$ (estimate from cell volume minus jelly-roll volume; refine by fitting).

$k_{vap}(T)$ follows Antoine equation behaviour for the electrolyte (typically EC/DMC/EMC mix); for engineering purposes you can approximate it as an exponential turn-on around 80°C and shut-off above 200°C once liquid electrolyte is exhausted.

3.2 Compute pressure

```
P = n_g * R * T / V_int    % R = 8.314 J/(mol·K), T in Kelvin
```

Subtract initial gas content so that $P(0) \approx 1 \text{ bar}$ matches measurement.

3.3 Identify the stoichiometric coefficients

v_{vap} , v_{e} , v_{O2} , V_{int} are unknown. Fit these to the pre-vent pressure curve of cell #1 only, then freeze.

Initial guesses based on chemistry:

- $v_{\text{vap}} = 1.0$ (each mole of liquid vapourises to one mole of gas)
- $v_{\text{e}} = 1.5$ (electrolyte decomposition produces CO_2 , CO , hydrocarbons — mix gives ~ 1.5 mol gas per mol electrolyte consumed)
- $v_{\text{O2}} = 0.5$ (typical for NMC811; depends on degree of delithiation)

3.4 Burst-disk model

Add a switch: if $P > P_{\text{burst}}$ (typically 25–35 bar for 21700), reset n_{g} to ambient-equivalent and trigger a "vented" flag. Pre-vent prediction is the primary contribution; post-vent is secondary.

Phase 4 — Validation and Sensitivity (Week 8)

Goal: generate the figures and tables that will go in the paper.

4.1 Validation runs

Run the frozen model on cells #2 and #3. Compute:

- RMSE on internal temperature (target: $<15^\circ\text{C}$ in pre-vent stage)
- RMSE on internal pressure (target: $<2 \text{ bar}$ in pre-vent stage)
- Time-to-onset error (target: <30 seconds)
- Soft-vent activation time error (target: <60 seconds)

If you cannot hit these targets, do not fudge it. Document the gap honestly in Section 6.2 of the paper. Reviewers respect honest limitations.

4.2 Sensitivity analysis

One-at-a-time sweeps with the frozen model:

- **SoC:** 50%, 60%, 70%, 80%, 90%, 100%. SoC affects: initial c_{ne} (anode lithium content), initial α (cathode delithiation), and v_{O2} (more oxygen release at higher SoC).
- **Heater power:** $\pm 25\%$ about experimental value.
- **Heat capacity:** $\pm 15\%$ about identified value (proxy for cell ageing / SoH variation).

Compute normalised sensitivity indices: $S_i = (\partial y / \partial x_i) \cdot (x_i / y)$ evaluated at the nominal point.

4.3 Produce paper figures

- Figure 1: model architecture (build in PowerPoint or Inkscape from screenshots of your Simulink subsystems)
- Figure 2: predicted vs measured internal temperature, two validation cells
- Figure 3: predicted vs measured internal pressure, two validation cells
- Figure 4: SoC sensitivity bar chart
- Figure 5: heater power and heat capacity sensitivity bar chart

All figures: vector format (PDF or SVG), serif font matching the paper, gridlines off, line widths ≥ 1.5 pt, font ≥ 10 pt.

Phase 5 — Writing and Submission (Weeks 9-10)

5.1 Fill in the draft

The Word draft I produced has placeholders like $[X]$, $[Y]$, $[Z]$. Fill these in with your actual numbers from Phase 4. Add Figures 1-5 and Tables 1-3.

5.2 Pre-submission checklist

- ☐ All claims in the paper map to figures, tables, or cited literature
- ☐ No claim relies on training data fits (only test/validation data)
- ☐ Code repository is public, with a frozen release tag matching the paper
- ☐ Dataset access is documented with a working DOI

- ☐ All co-authors agree (if it's only you, list yourself with affiliation)
- ☐ You have read the target journal's author guidelines and formatted accordingly

5.3 Target journals (descending ambition)

1. **Journal of Energy Storage** (Elsevier, IF ~9) — where Marco's group publishes; competitive but realistic if the work is solid
2. **Journal of Power Sources Advances** (Elsevier, open access) — also where the group publishes, easier acceptance for novel methodologies
3. **Batteries** (MDPI, IF ~5) — open access, faster review, good fallback
4. **Energies** (MDPI, IF ~3) — fastest, most lenient, weakest signal of quality

I recommend submitting to (2) first. Open access is a strong signal to Marco that you wanted maximum visibility, and the journal is in the group's citation network.

5.4 The email to Marco — timing matters

Do not email Marco until:

- The paper is submitted (not before — premature contact looks like you wanted a co-author rather than wanted to do the work)
- You have a preprint posted on arXiv or ChemRxiv

Then the email looks like this (rough sketch):

Dear Prof. Marco,

Following our earlier discussion about the PhD opportunity at WMG, I wanted to share work I have completed independently in the intervening period. I have developed a coupled electro-thermal-pressure model for prediction of internal gas pressure during thermal runaway in 21700 cells, validated against the open-access dataset published by Gulsoy et al. earlier this year. The preprint is at [link] and the open-source code is at [link]; the paper is under review at [journal].

The work was motivated by what I saw as a gap between the experimental capabilities your group has demonstrated and the reduced-order modelling literature, where internal pressure prediction has been comparatively neglected. I would value any feedback you have, and would be very interested if any future PhD projects in your group — including industry-sponsored or self-funded routes — align with this direction.

[Signature]

Short. Concrete. Demonstrates independent capability. Asks for nothing he cannot

Short. Concrete. Demonstrates independent capability. Asks for nothing he cannot give. Mentions self-funded explicitly because that signals you understand the funding constraint and have thought about workarounds.

Honest checkpoints — when to stop and reconsider

This plan is realistic but ambitious. At each phase, ask yourself:

- **End of Phase 1:** can I reproduce a published TR curve to within 50 °C peak temperature? If no, the modelling work is beyond your current toolset and you should pivot to a less mathematical contribution (e.g. literature review paper on TR detection algorithms — much easier to publish).
- **End of Phase 2:** does the fitted model match the training cell to within 30 °C RMSE? If no, your parameter identification is broken — usually means the optimiser is escaping the physical regime. Tighten bounds.
- **End of Phase 3:** does the pressure prediction at least follow the right shape (slow rise, then acceleration to burst)? If no, your gas-generation stoichiometry is wrong or V_{int} is wildly off.
- **End of Phase 4:** is the validation RMSE in the same order as the training RMSE? If validation is much worse, you have overfitted.

If any checkpoint fails and you cannot diagnose it within a week, that is the signal to **simplify the contribution** rather than push forward with a broken model. A simpler paper that is correct is infinitely better than an ambitious paper that gets rejected for unphysical parameters.

Time-on-task estimate, honestly

If you have full-time hours: 8–10 weeks.

If you have evenings and weekends only: 4–6 months.

There is no shortcut. The paper I sketched is what a careful researcher with relevant background would produce in roughly that time. If you try to compress it, the work will show, and Marco will see that immediately.

Good luck. This is a hard route but it's not an unreasonable one.